

CDAOs agreed that effective data and analytics governance is essential for business and technology innovation.

Challenges in data governance

Enterprises today face a range of data governance challenges that hinder effective data utilisation and decision-making. One of the most significant issues is data being stored in silos—scattered across platforms, tools, channels, and business units—which makes access difficult and leads to inefficiencies and inconsistencies. Ensuring data accuracy, completeness, and timeliness remains a persistent struggle. Additionally, the lack of oversight in monitoring both incoming data quality and its usage across the organisation results in degraded data value. Converting offline data into usable digital formats poses another hurdle. Enterprises also grapple with managing regulatory compliance while trying to maintain data security and privacy.

A poorly defined data management strategy can lead to overwhelming volumes of disorganised and unmanageable data. This, in turn, increases the risk of sensitive business or customer information being leaked or misused. Furthermore, the use of unsecured data from various sources opens the door to data breaches. The complexity is compounded by the adoption of modern digital applications that rely on polyglot storage architectures—such as relational databases, columnar databases, key-value stores, and graph databases—which vary depending on application functionality, making unified governance even more challenging.

Next-gen data management capabilities

To manage the huge and different types of data being generated today, businesses need to upgrade their data management capabilities.

Data processing techniques:

Unstructured data processing covers entity extraction, concept extraction, sentiment analysis, NLP, ontology, etc. Machine learning techniques must be leveraged to automate this extraction.

Agile data delivery: The overall cost of data delivery can be reduced by using agile development methodology.

Digital 360 view of the customer and products:

Customers interact with enterprises through social media channels. The ability to create a single view of a customer and the company's products across internal and external data is critical in the digital world.

On-demand data services: It's important for businesses to provide virtualised access to data by way of on-demand data services for both online and offline use.

Automated metadata extraction:

Data is thrown into enterprise systems at a rate that doesn't allow businesses (especially SMEs) to look at data structures and extract metadata. Automated metadata extraction based on ontology is critical.

Real-time analytics: Businesses need to use multi-channel applications and decision management systems to capture the interactions of digital processes in real-time scenarios.

Data archival: Compliance and performance requirements drive the need for archival of both structured and unstructured data.

Principles of data governance

Data governance principles are a structured set of ideas that collectively define and guide the development of a solution architecture, from values through to design and implementation, harmonising decision-making across an organisation.

These include:

- **Data is an asset:** Data is an asset that has a specific and measurable value for an enterprise.
- **Data is shared:** Users have access to the data necessary to perform

their tasks; therefore, data is shared across enterprise functions and organisations.

- **Data trustee:** Each data element has a trustee accountable for data quality.
- **Common vocabulary and data definitions:** Data definition is consistent throughout the enterprise, and the definitions are understandable and available to all users.
- **Data security:** Data is protected from unauthorised use and disclosure.
- **Data privacy:** Privacy and data protection is enabled throughout the life cycle of the data. All data sharing conforms to relevant regulatory and business requirements.
- **Data integrity:** Each enterprise must be aware of and abide by its responsibilities with respect to the provision of source data and the obligation to establish and maintain adequate controls over the use of personal or other sensitive data.
- **Data transparency:** Data governance processes must be transparent. All data-related decisions must be explained clearly to all personnel as to how, when, and why they were introduced.
- **Fit for purpose:** A next-generation information ecosystem needs to have fit for purpose tools since one technology will not meet the needs of all the workloads and processing techniques. These include text processing, data discovery, dynamic data services, high performance analysis, streaming analytics, etc.

Key components of data governance

Data governance ensures data quality, accuracy, and compliance with regulatory requirements, building trust in the data. It must be aligned with enterprise architecture governance to realise the objectives of the business.